Exercises: Probability — Basics Mathematical Foundations for Machine Learning

Discrete Probability and Distributions

Exercise 1.1 [Easy]

A fair six-sided die is rolled once. Let X be the random variable defined as

$$X = \begin{cases} 1 & \text{if the outcome is even,} \\ 0 & \text{otherwise.} \end{cases}$$

(a) Find the probability mass function P_X.

(b) Compute $\mathbb{E}[X]$ and $\mathrm{Var}[X]$.

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k}$$

binomial theorem

Exercise 1.2 [Medium]

A biased coin has probability p of showing heads. It is tossed n times independently. Let X denote the number of heads observed.

(a) Write down the probability mass function of X and verify that the probabilities sum to 1.

1- k = 0 is useless

(b) Using the binomial distribution, show that $\mathbb{E}[X] = np$.

2- substract n-1 and change to j = k-1

(c) A machine learning classifier makes independent predictions on 20 test samples, each correct with probability 0.85. What is the probability that exactly 17 predictions are correct?

Exercise 1.3 [Hard]

Consider a discrete probability space $(\Omega, 2^\Omega, P)$ with $\Omega = \{\omega_1, \omega_2, \omega_3\}$ and $P(\omega_i) = p_i > 0$ for $i = 1, 2, 3$.

(a) How many events does 2^Ω contain? Verify that the additivity rule $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ holds for $A = \{\omega_1, \omega_2\}$ and $B = \{\omega_2, \omega_3\}$.

(b) Define random variables $X, Y : \Omega \to \mathbb{R}$ by $X(\omega_i) = i$ and $Y(\omega_i) = i^2$. Compute $\mathbb{E}[XY]$ in terms of p_1, p_2, p_3.

(c) When moving from a finite Ω to an uncountable one like $\mathbb{R}$, the powerset 2^Ω must be replaced by a σ-algebra. Give the three defining properties of a σ-algebra and briefly explain why this restriction is necessary.

Continuous Distributions and Density Functions

Exercise 2.1 [Easy]

Consider the function

$$p(x) = \begin{cases} c\,(1 - x^2) & \text{if } x \in [-1, 1], \\ 0 & \text{otherwise.} \end{cases}$$

(a) Find the constant $c > 0$ such that p is a valid probability density.

(b) Compute the cumulative distribution function $F(x) = P((-\infty, x])$ for $x \in [-1, 1]$.

Exercise 2.2 [Medium]

cumulative distribution -> count from x to the most negative

Let $X \sim \mathcal{N}(\mu, \sigma^2)$ be a univariate Gaussian random variable with density

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

$$p(\mu + t) = p(\mu - t) \quad \text{for all } t \in \mathbb{R}.$$

-> PDF is a symmetric curve

$$-\int_b^a f(x)\,dx = \int_a^b f(x)\,dx$$

median -> P (X m) = 1/2 -> change of variable, definition of symmetric -> sum = 1

(a) Show that if p is symmetric around μ, then the mean and the median of X coincide.

(b) The α-quantile q_α satisfies $F(q_\alpha) = \alpha$. Using symmetry of p around μ, show that $q_\alpha = 2\mu - q_{1-\alpha}$.

(c) In a machine learning context, a model's prediction error follows $\mathcal{N}(0, 4)$. Find an expression (in terms of q_α) for the symmetric interval $[-a, a]$ that contains 95% of the errors.

$$P(Z \le q_\alpha^{\text{std}}) = \alpha$$

quatile when N(0,1)

$$P\left(Z \le \frac{q_\alpha - \mu}{\sigma}\right) = \alpha$$

substracting u and div by stnd.

$$\frac{q_\alpha - \mu}{\sigma} = q_\alpha^{\text{std}}$$

then

Exercise 2.3 [Hard]

Let $X \sim \mathcal{N}(\mu, C)$ be a $\mathbb{R}^d$-valued random variable with density

$$p(x) = \frac{1}{\sqrt{(2\pi)^d \det C}} \exp\left(-\frac{1}{2}(x-\mu)^T C^{-1}(x-\mu)\right),$$

where $C \in \mathbb{R}^{d\times d}$ is symmetric and positive definite.

(a) Show that the density $p(x)$ is constant on the ellipsoid $\{x \in \mathbb{R}^d : (x-\mu)^T C^{-1}(x-\mu) = r^2\}$ for any fixed $r > 0$.

(b) Let $C = \sigma^2 I$ (isotropic case). Show that the density depends on x only through $\|x-\mu\|_2$, and hence the level sets are spheres.

(c) Let $v \in \mathbb{R}^d$ be a fixed unit vector. Define the scalar random variable $Z = v^T X$. Using the properties of expectation and the covariance matrix, show that $\mathbb{E}[Z] = v^T\mu$ and $\mathrm{Var}(Z) = v^T C v$. Use tr for Var(Z)!

(b): Since covariance matrix says all vars same variance + no correlation between vars, the ellipsoid that represents the levelset in t

Joint Distributions, Marginals, and Transformations

Exercise 3.1 [Easy]

$$\mathrm{Vol}(T(S)) = |\det(A)| \cdot \mathrm{Vol}(S)$$

Let (X_1, X_2) be a random variable in $\mathbb{R}^2$ with joint density

$$p(x_1, x_2) = \begin{cases} 6\,x_1\,x_2^2 & \text{if } 0 < x_1 < 1,\ 0 < x_2 < 1, \\ 0 & \text{otherwise.} \end{cases}$$

volume means the area of the set in which we measure probabilities
= integral(1)dx -> when p(x) = 1 always
The det of trans. says how scale volume

(a) Verify that p is a valid joint density.

(b) Compute the marginal densities $p_{X_1}(x_1)$ and $p_{X_2}(x_2)$.

Exercise 3.2 [Medium]

$$P(y \le Y \le y + dy) \approx p_Y(y) \cdot dy$$

density * interval = prob of being inside
density = probability per unit interval

Suppose U is uniformly distributed on $[0, 1]$, i.e. $p_U(u) = 1$ for $u \in [0, 1]$.

(a) Using the transformation law, find the density of $Y = -\frac{1}{\lambda}\log(U)$ for $\lambda > 0$. What well-known distribution does Y follow?

(b) Now let $Z = U^2$. Using the transformation law, find the density $p_Z(z)$.

(c) In the context of sampling: suppose you only have access to a uniform random number generator. Explain how you would use part (a) to generate samples from an exponential distribution with rate $\lambda = 2$.

$$P(a \le Y \le b) = \int_a^b p_Y(y)\,dy$$

therefore you can integrate to take the prob. of being inside an interval

Exercise 3.3 [Hard]

Let X be a $\mathbb{R}^d$-valued random variable with density p_X, and let $g : \mathbb{R}^d \to \mathbb{R}^d$ be a continuously differentiable bijection with non-vanishing Jacobian determinant.

(a) State the transformation law: write the density p_Y of $Y = g(X)$ in terms of p_X, g^{-1}, and the Jacobian of g^{-1}.

(b) Let $d = 2$, $X \sim \mathcal{N}(0, I_2)$, and define $g(x_1, x_2) = (2x_1 + x_2,\ x_2)$. Compute the Jacobian matrix $Dg(x)$, its determinant, and find the density $p_Y(y)$.

$$\frac{d}{dy}F_Y(y) = p_Y(y)$$

$$F_Y(y) = P(Y \le y)$$

$$F_Y(y) = P(g(U) \le y) = P(U \le g^{-1}(y)) = F_U(g^{-1}(y)) \quad p_Y(y) = \frac{d}{dy}F_U(g^{-1}(y)) = p_U(g^{-1}(y)) \cdot \frac{dg^{-1}}{dy}$$

assume monotonically increasing

bec. monoton. inc. then preserv. ineq. dir.

$$Y = g(U)$$

Pick U with unif. density = 1

$$p_Y(y) = p_U(g^{-1}(y)) \cdot \left|\frac{dg^{-1}}{dy}\right|$$

we take abs. value because if not increasing, then decreasing (else not inverse and when derivating, the derivate of pU is already negative so positive, it is th

(c) In the setting of **(b)**, is Y still Gaussian? If so, find its mean and covariance matrix C_Y directly from the linear transformation $Y = AX$ and verify it matches the density you found.

Expectation, Variance, Covariance, and Correlation

Exercise 4.1 [Easy]

Let X and Y be two real-valued random variables with $\mathbb{E}[X] = 2$, $\mathbb{E}[Y] = -1$, $\mathrm{Var}(X) = 3$, $\mathrm{Var}(Y) = 5$, and $\mathrm{Cov}(X, Y) = -2$.

(a) Compute $\mathrm{Var}(X + Y)$ and $\mathrm{Var}(2X - 3Y + 7)$.

(b) Compute $\mathrm{Corr}(X, Y)$ and determine whether X and Y are positively or negatively correlated.

Exercise 4.2 [Medium]

A coin has probability p for heads. It is tossed twice independently. Let X denote the result of the first toss and Y the result of the second toss (1 for heads, 0 otherwise).

(a) Compute $\mathbb{E}[XY]$. *(cf. Exam 2021, Ex. 4b)*

(b) Compute $\mathrm{Cov}(X, Y)$ and explain the result using independence.

(c) Now define $Z = X + Y$ (total number of heads). Compute $\mathrm{Cov}(X, Z)$ and $\mathrm{Corr}(X, Z)$.

Exercise 4.3 [Hard]

Let X be a real-valued random variable with a symmetric density, i.e. $p(x) = p(-x)$ for all x, and let $Y = X^2$.

(a) Show that $\mathrm{Cov}(X, Y) = 0$.

(b) Despite zero covariance, X and Y are clearly dependent (since Y is a function of X). Does $\mathrm{Corr}(X, Y) = 0$ imply independence? Justify your answer.

(c) Now let $X \sim \mathcal{N}(0, \sigma^2 I_d)$ be a d-dimensional Gaussian and $A \in \mathbb{R}^{d\times d}$ symmetric. Show that

$$\mathbb{E}[\langle X, AX\rangle] = \sigma^2 \,\mathrm{tr}(A).$$

Independence, Conditional Probability, and Bayes' Theorem

Exercise 5.1 [Easy]

Let (X_1, X_2) have joint density $p(x_1, x_2) = 6\, x_1\, x_2^2$ on $[0, 1]^2$ (as in Exercise 3.1).

(a) Using the marginal densities $p_{X_1}(x_1) = 2x_1$ and $p_{X_2}(x_2) = 3x_2^2$, determine whether X_1 and X_2 are independent.

(b) Compute $\mathbb{E}[X_1 X_2]$ and verify that it equals $\mathbb{E}[X_1]\,\mathbb{E}[X_2]$.

Exercise 5.2 [Medium]

A medical test for a disease has the following properties: if a patient has the disease, the test is positive with probability 0.95 (sensitivity). If the patient is healthy, the test is positive with probability 0.03 (false positive rate). The prevalence of the disease in the population is 0.01.

(a) Using Bayes' theorem, compute the probability that a patient actually has the disease given a positive test result.

(b) The result of part (a) is often surprisingly low. Using the law of total probability, explain intuitively why this happens by computing $P(\text{positive test})$.

(c) At what prevalence level π would a positive test result give a 50% probability of having the disease? *(cf. Exam 2021, Ex. 4c)*

Exercise 5.3 [Hard]

Let X and Y be real-valued random variables with joint density $p(x, y)$.

(a) Write down the conditional density $p(x|y)$ and verify that $\int_{\mathbb{R}} p(x|y)\, dx = 1$ for any y with $p(y) > 0$.

(b) Prove the tower property of conditional expectation: $\mathbb{E}[\mathbb{E}[X|Y]] = \mathbb{E}[X]$.

(c) In a mixture of discrete and continuous spaces: let $Y \in \{1, \ldots, K\}$ be a class label and $X \in \mathbb{R}^d$ a feature vector with joint distribution $p(x, k)$. Show that the posterior $P(Y = k|x) = \frac{p(x|Y=k)\, P(Y=k)}{\sum_{l=1}^{K} p(x|Y=l)\, P(Y=l)}$ defines a valid probability measure over $\{1, \ldots, K\}$ for each x.

Mixed Challenging Exercises

Exercise 6.1 [Hard]

Consider N data points $\{(x_n, y_n)\}_{n=1}^{N}$ with $x_n \in \mathbb{R}^D$ and $y_n \in \mathbb{R}$, generated by the model $y = x^T\theta + \varepsilon$ where $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ i.i.d.

(a) Write the conditional density $p(y_n|x_n; \theta)$ for a single observation.

(b) Using the i.i.d. assumption, write the joint conditional density $P(Y|X; \theta)$ for all N observations (where $X \in \mathbb{R}^{N\times D}$ is the design matrix and $Y \in \mathbb{R}^N$ the target vector).

(c) Show that the maximum likelihood estimate of θ is $\hat{\theta} = (X^TX)^{-1}X^TY$.

(d) Show that maximizing the log-likelihood is equivalent to minimizing $\|Y - X\theta\|_2^2$. Explain the connection: under what assumption on the noise does least-squares regression correspond to maximum likelihood?

(cf. Exam 2021, Ex. 4d)

Exercise 6.2 [Hard]

Let $X = (X_1, \ldots, X_d)$ be a random vector with $X_i \sim \text{Uniform}(0, 1)$ independently.

(a) Compute $\mathbb{E}[X_i]$ and $\text{Var}(X_i)$ for each component.

(b) Let $S = \sum_{i=1}^{d} X_i$. Using the properties of expectation and variance for independent random variables, find $\mathbb{E}[S]$ and $\text{Var}(S)$.

(c) Apply the Chebyshev inequality to bound $P\big(|S - \mathbb{E}[S]| \geq t\big)$ for any $t > 0$.

(d) For what value of d can you guarantee, via Chebyshev, that $P\big(\big|\frac{S}{d} - \frac{1}{2}\big| \geq 0.1\big) \leq 0.05$?

Exercise 6.3 [Hard]

Consider a classification model that assigns class labels $Y \in \{1, 2, 3\}$ to inputs $x \in \mathbb{R}^d$. The class-conditional densities are Gaussian: $p(x|Y = k) = \mathcal{N}(x; \mu_k, \Sigma)$ with equal covariance Σ and equal prior probabilities $P(Y = k) = \frac{1}{3}$.

(a) Using Bayes' theorem, write the posterior $P(Y = k|x)$ and simplify using the equal prior assumption.

(b) Show that the decision boundary between classes k and l (where $P(Y = k|x) = P(Y = l|x)$) is a hyperplane. Find the equation of this hyperplane in terms of μ_k, μ_l, and Σ.

(c) Is the MAP (maximum a posteriori) decision rule the same as the MLE decision rule when priors are equal? Justify.